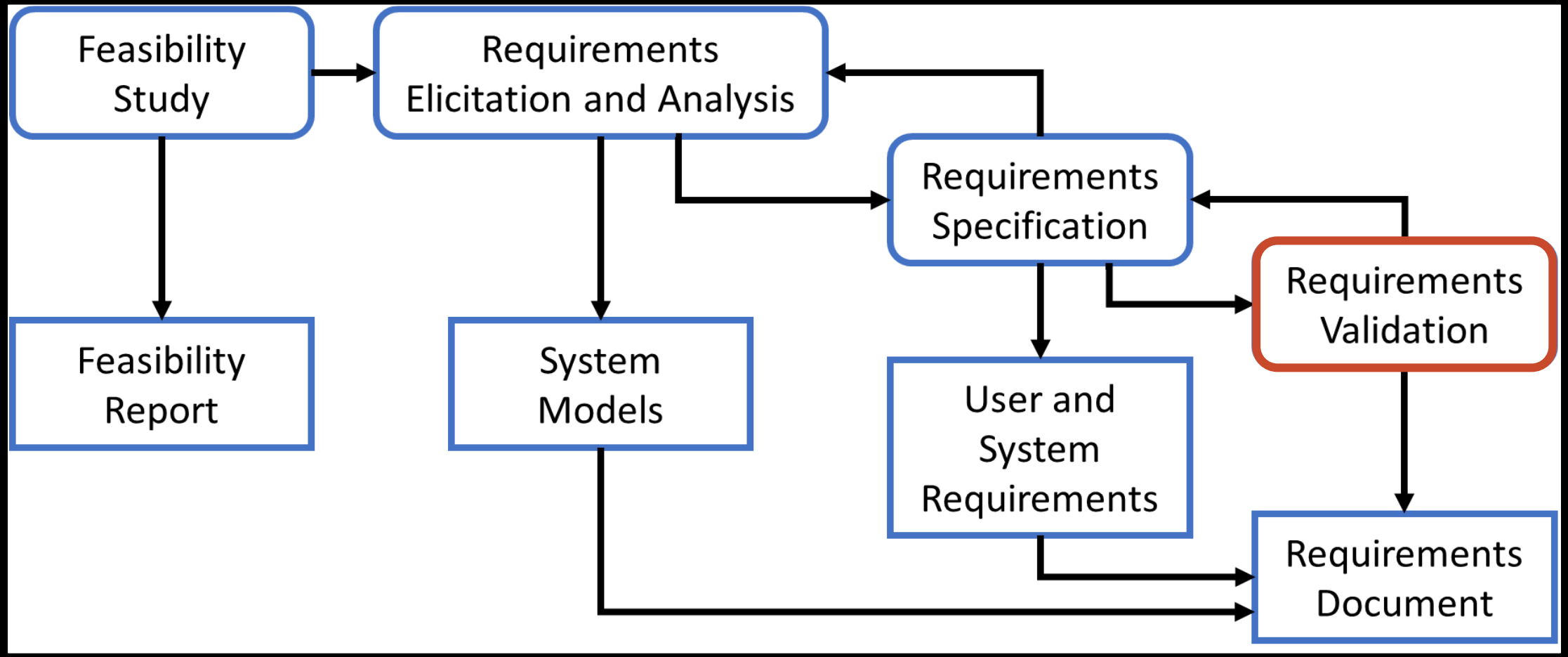
Software Engineering Management COMP4135

Seminar 5

Prototyping

Requirement Validation



Defining Prototype Goals

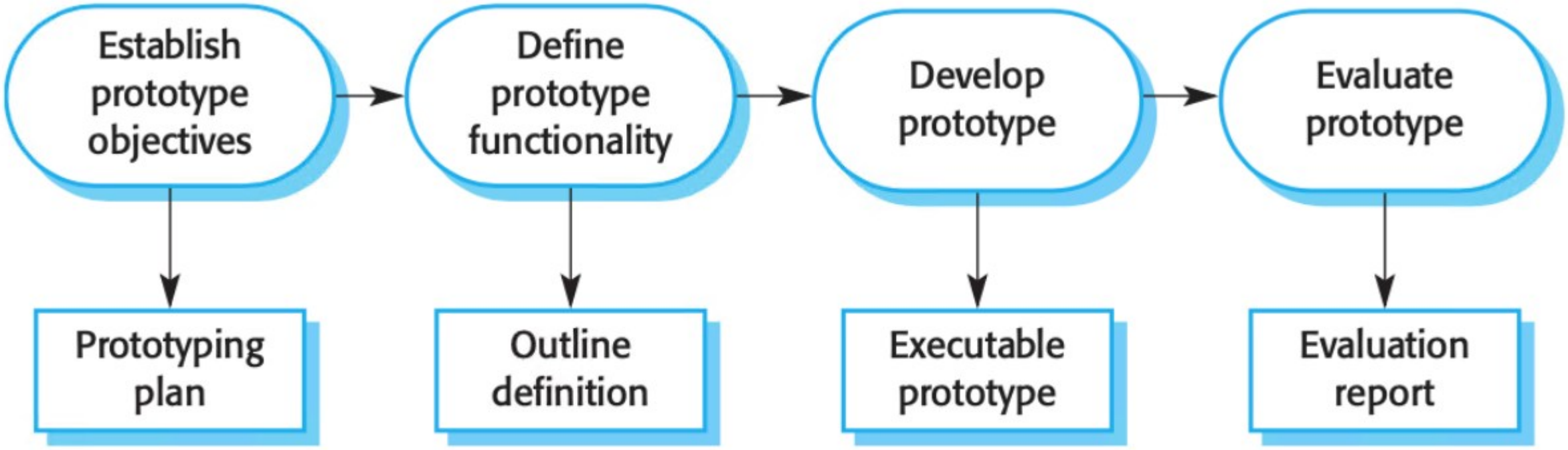
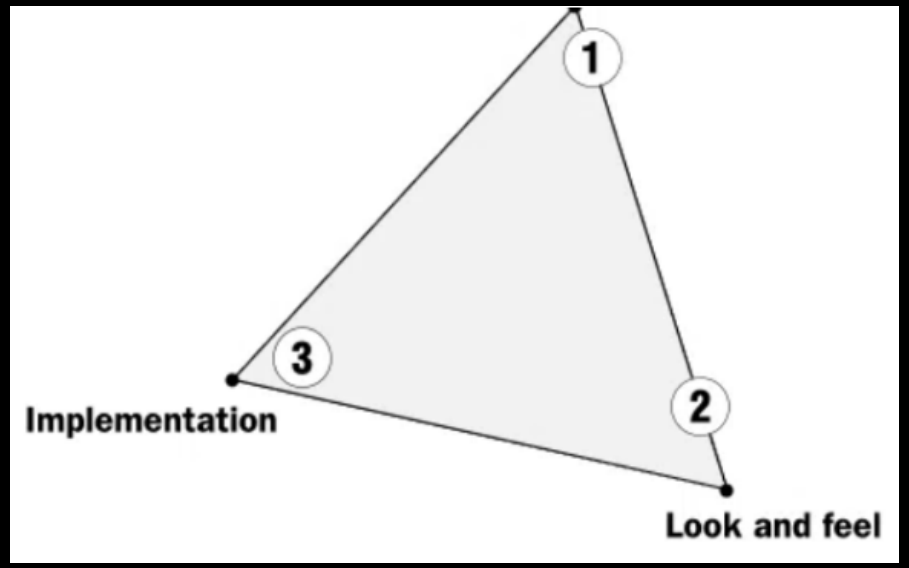


Fig: Prototype Development

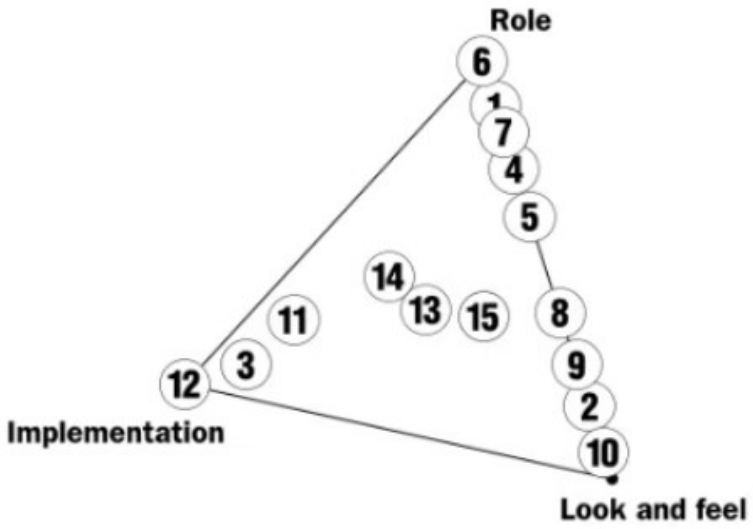
3 Purposes of Prototype

- Prototypes fill a purpose.
  1. Role of technology.
  2. Look and Feel.
  3. Implementation Guide.
- UI Designers do #2.
- Powerpoint or paper prototypes do more #1.
- In some companies
  - Aimed at all employees (company vision).
  - Does #1 – Does Not Do #2 or #3.



Prototypes >>Implementation

- Prototypes help everyone to image what we are building.
- The customer, the manager, the developer etc.
- But we still need to be able to tell a developer.
- You go and build X.
- Look & Feel does it for UI Devs.
- Powerpoints could guide Implementation.
- Sometimes we code up prototypes.
- These should be partial implementations.

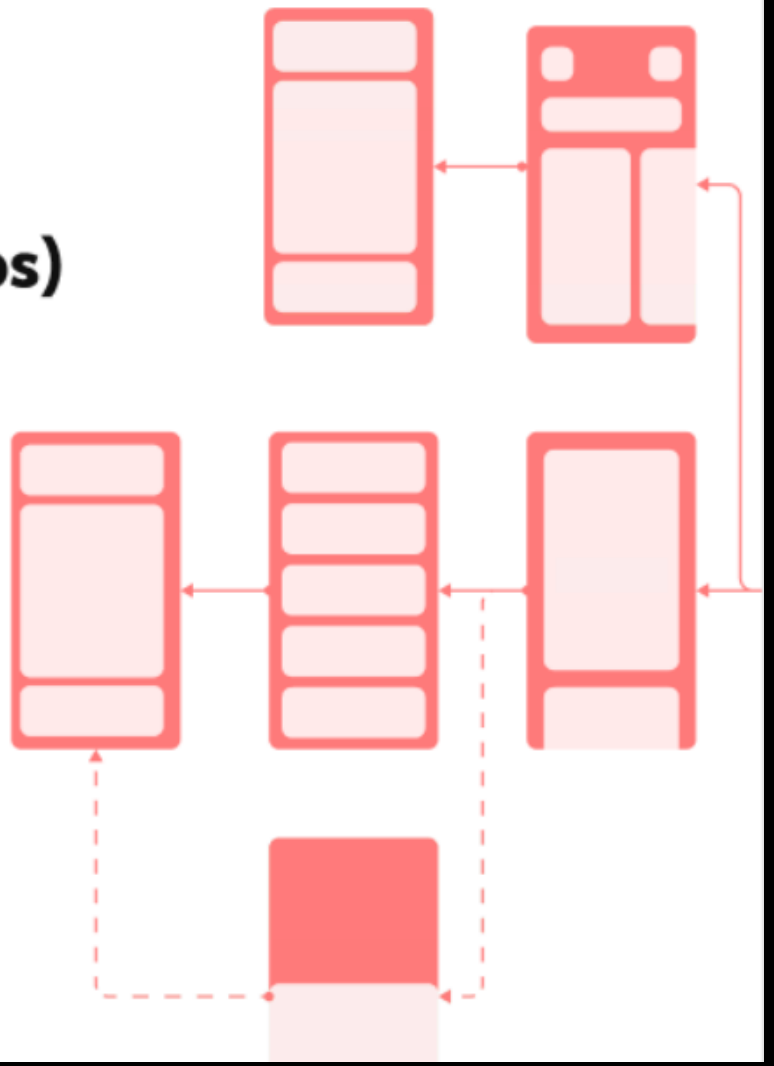


1. *3D space-planning (role)*
2. *3D space-planning (look and feel)*
3. *3D space-planning (implementation)*
4. *Storyboard for portable notebook computer*
5. *Interactive story, operating system user interface*
6. *Vision video, notebook computer*
7. *Appearance model, integrated communicator*
8. *Animation, fashion design workspace*
9. *Look and feel simulation, child's toy*
10. *Pizza-box, architect's computer*
11. *Working prototype, digital movie editor*
12. *C++ program listing, fluid dynamics simulation*
13. *Integrated prototype, sound browser*
14. *Integrated prototype, pile metaphor*
15. *Integrated prototype, garment history browser*

Prototype

10 Types of Prototypes (With Explanations and Tips)

- Sketches and Diagrams
- Wireframe
- 3D Printing Prototype
- Physical Model
- VR & AR Prototypes
- Feasibility
- Video Prototype
- Horizontal Prototype
- Fully Functional Rapid Prototype



Prototype vs Model

	Prototype	Model
What	A preliminary version of a product or system created for testing and evaluation	A representation or simulation of a system used to understand or predict its behavior.
How	Typically, physical or functional in nature.	Can be physical or virtual.
Focus	Focused on testing and refining a specific product or system.	Focused on understanding and predicting the behavior of a broader system or process.
Purpose	Used to identify and fix design issues.	Used to explore different scenarios and make prediction
Format	May not be fully functional or representative of the final product or system.	Designed to accurately represent the system being modeled.
Stage	Created early in the development process.	Created at any stage of the development process.
Cost	Can be expensive to produce.	Can be less expensive to produce than physical prototypes.

Prototyping Risks

1. Investing too much time/energy on high-fidelity prototypes.
   - When a low fidelity one would let you test the point.

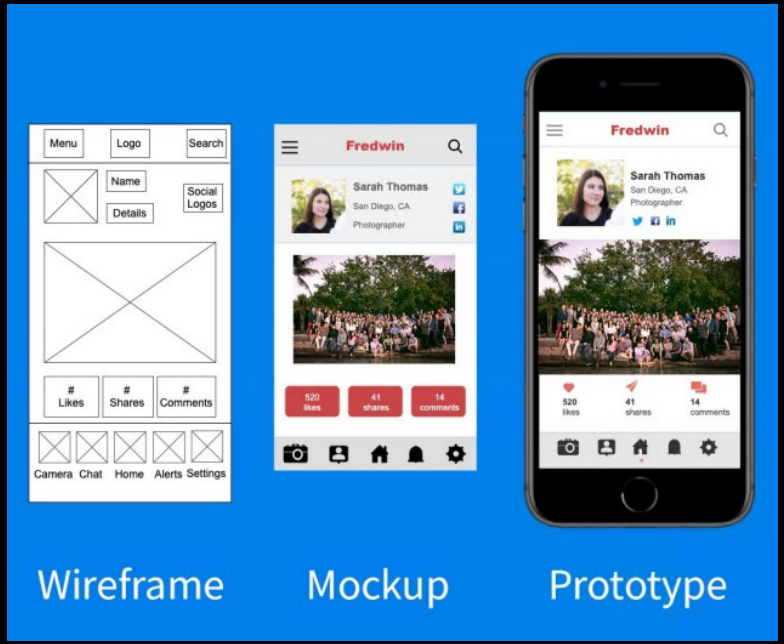
2. Adhoc prototyping code is re-used in the real system.
   - But the code wasn't produced to a professional standard.

3. Prototying is used instead of, rather than alongside, documentation.
   - Which may then be insufficient for software maintenance phase.

4. Prototypes might be approved by the wrong stakeholders.
   - E.g., managers rather than the end users.

Wireframe vs Mockup vs Code Prototype

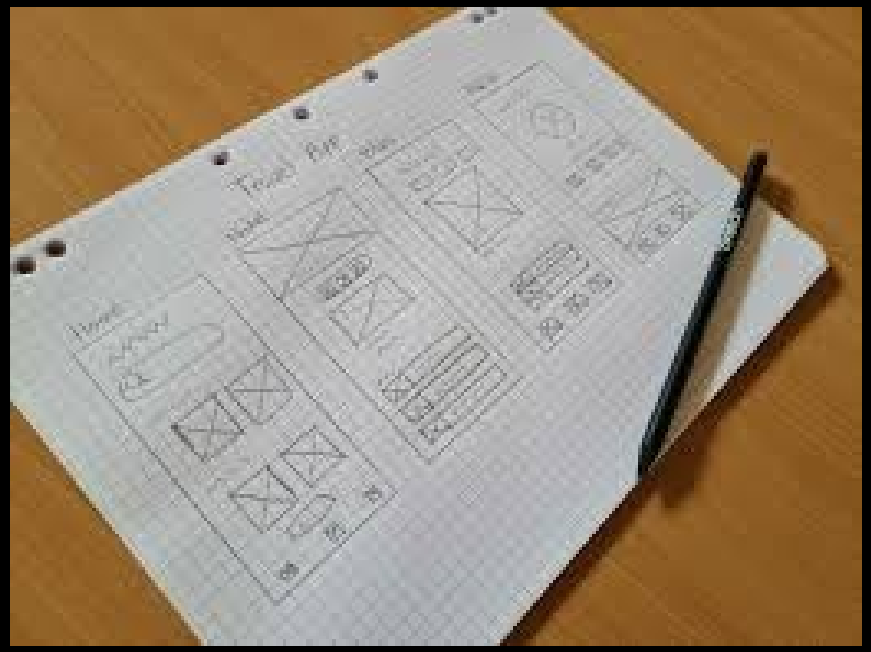
	Wireframe	Mockup	Prototype
What	A quick sketch to convey the high-level concept of new product functionality	A realistic visual design that resembles what the new product functionality will look like	Interactive simulation of new product functionality
Purpose	To gain consensus and collect internal feedback on how new functionality will work	To facilitate more detailed critiques of visual elements and functionality so changes can be made	To collect feedback by user testing the real experience
Included elements	The format and structure of content	Additional visual elements like logos, colors, and icons	Final interactive elements and navigation
Creator	PM or UX Designer	UX Designer	UX Designer
Design fidelity	Low	Medium	High



Group Discussion – Design your Game

Think about your weekly report1 – Requirement Engineering:

- Consider the basic design of your game
- In your team, draw a wireframe for your team



Prototyping Tools

- Paper prototyping templates.
  - E.g., https://sketchsheets.com/

- Many good prototypes can be built in e.g., PowerPoint.
  - Can be semi-interactive!
  - Buttons link to different slides.

- Apps to help prototyping.
  - https://marvelapp.com/
  - https://www.figma.com/
  - https://proto.io/en/demos/
  - Hi-Fi tools e.g., https://www.sketch.com/

Assessment !!! Week Report 2

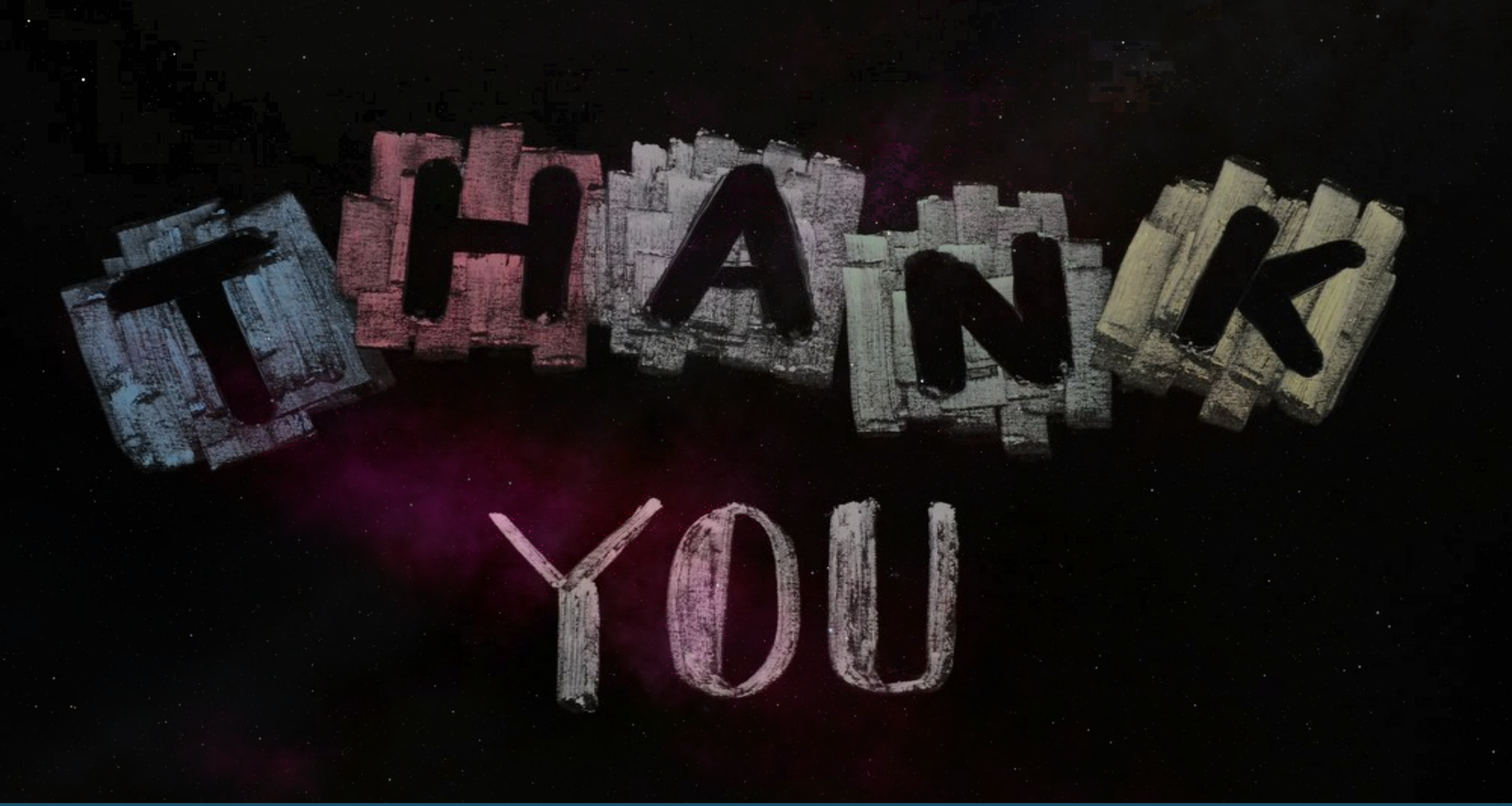
Your group is in a software company. You already have an requirement engineeri

- You should think about the requirement changing due to the conflicts, so you can discuss them. (requirement validation)
- You should think about the basic design and interactions with users, so you can discuss them. (prototype)
- Think about the implementation strategies.
- You can define the requirements and specification better. To make good content, consider the following:
  - Producing basic diagrams.
  - Producing good diagrams that are useful.
  - Explaining reasoning for choices and useful insights.

Assessment !!! Week Report 2

Reminder:

- Submission Deadline: 3 April 2026, 17:00
- Follow the template on Moodle
- Only team leader need to submit the team report
- The team report link should also shown in your Team GitLab


THANK
YOU